CAD MODULE

Session 1 · Floor Plan

Objectives

1. Set up the drawing environment and build a layer system.　建立绘图环境与图层系统。
2. Draw an axis grid with typed offset distances.　用键入的偏移距离绘制轴网。
3. Draw 240 mm walls with multiline and clean the joints.　用多线绘制 240 墙并编辑交点。
4. Hatch what the section cut passes through.　对被剖切到的实体进行填充。

In-class exercise

Draw the ground-floor plan of the villa from the issued base drawing: the axis grid, the exterior wall, and the hatching of columns and paving.

1　绘图环境　Drawing Environment

1.1　显示设置　Display settings

OPTIONS ↲　right-click in the drawing area → Options → Display　右键 → 选项 → 显示

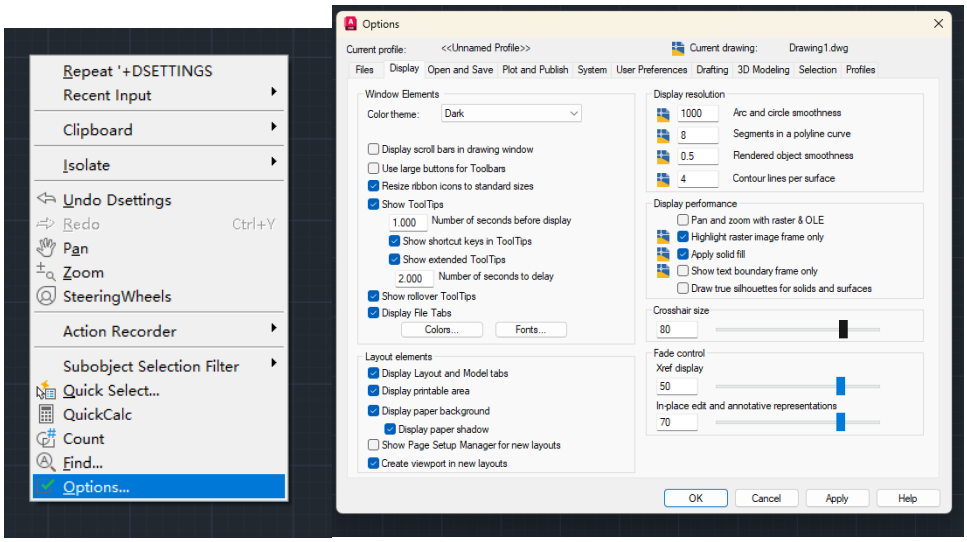
1. Right-click on an empty part of the drawing area.　在绘图区空白处点右键。
2. Choose Options… at the bottom of the menu.　选择菜单最下方的 Options。
3. Display tab → Crosshair size = 80.　十字光标大小设为 80。

※ A crosshair at 80 spans the screen and works as an alignment guide: you can see whether two points are level without drawing a construction line.

1.2　拾取框与夹点　Pickbox and grips

OPTIONS ↲　Options → Selection tab　选项 → 选择集

1. Options → Selection tab.
2. Pickbox size — move the slider to the middle.
3. Test it: draw a line, then try to click on it. If you miss, go back and enlarge the pickbox.

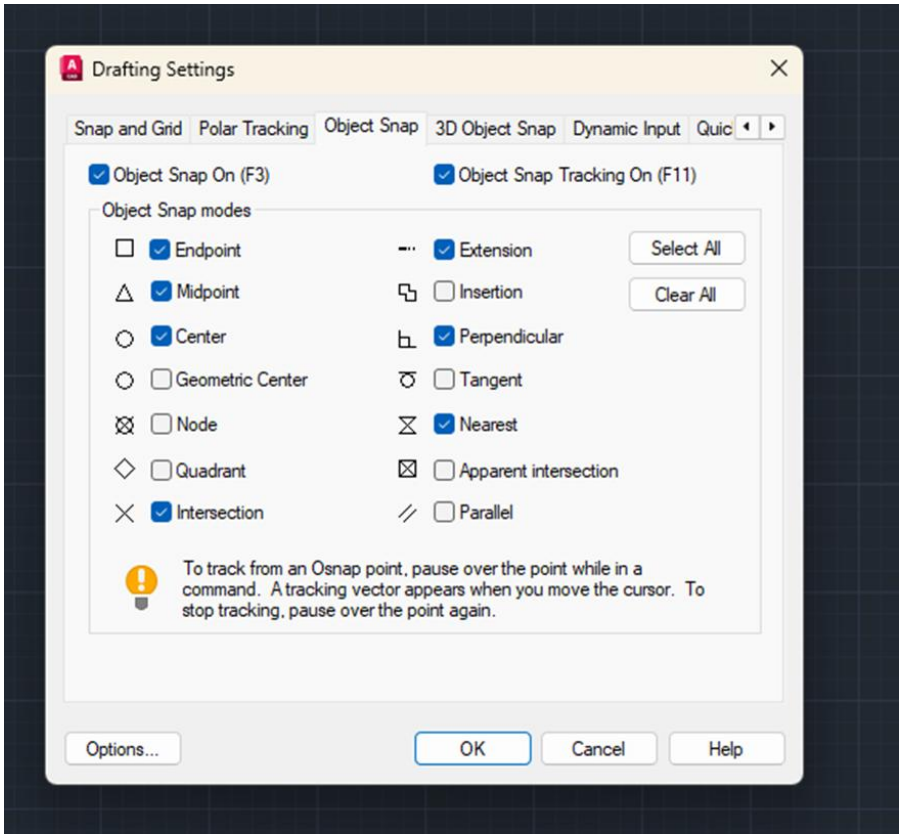


1.3　对象捕捉　Object snap

DSETTINGS ↵ （DS ↵） Tools → Drafting Settings → Object Snap 工具 → 绘图设置 → 对象捕捉

Snap 捕捉	Snaps to 捕捉对象
Endpoint	the end of a line
Midpoint	the middle of a line
Intersection	where two lines cross — used for every axis intersection
Perpendicular	a right-angle foot on another line

※ Ticking every snap makes the cursor jump to the wrong point constantly. Four is enough for a floor plan.
全部勾选会导致光标不断捕捉到错误的点。



1.4 功能键 Function keys

Key	Function 功能	Use 用途
F3	Object Snap 对象捕捉	Off = every point you click is only approximately right.
F8	Ortho 正交	Locks the cursor to horizontal / vertical. Keep it ON all session.
F10	Polar Tracking 极轴	Tracks set angles. Not needed today.
F12	Dynamic Input 动态输入	Shows the value box at the cursor.

2 图层 Layers

2.1 图层管理器 Layer Properties Manager

LAYER ↵ （LA ↵） Format → Layer 格式 → 图层

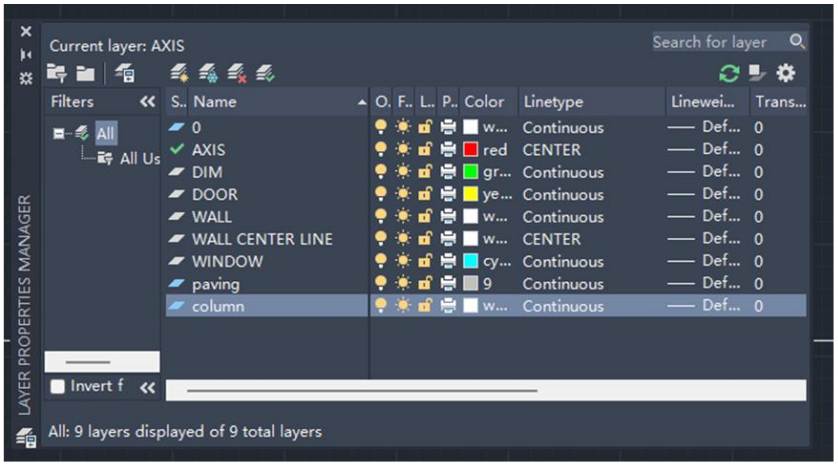
1. New layer — click the New Layer icon or press Alt + N. 新建图层：Alt + N。

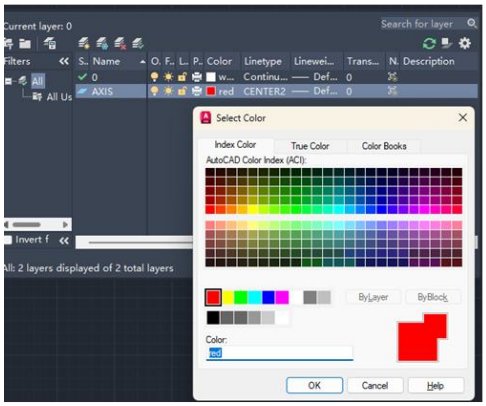
2. Delete layer — select it, then Alt + D.　删除图层：Alt + D。
3. Rename — click the name once and type. Name the layer after the element it carries.
4. Set colour, linetype and lineweight in the corresponding columns.
5. Double-click a layer name to make it the current layer. New objects land on the current layer.

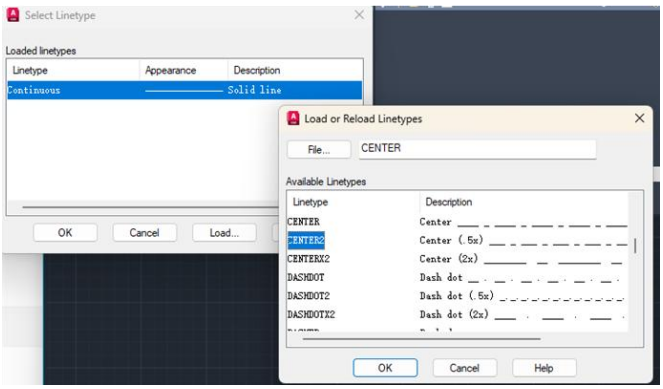
※ Never draw on layer 0. Layer 0 is only a container for block definitions.　不要在图层 0 上绘图，它只作为图块定义的容器。

2.2　七个图层　The seven layers

Build all seven before you draw anything.　绘图之前先把七个图层全部建好。







2.3　线型与线型比例　Linetype and LTSCALE

LINETYPE ↲ · LTSCALE ↲ · REGEN ↲　Format → Linetype　格式 → 线型

1. LINETYPE ↲ → Load… → select CENTER2 → OK. CENTER2 is not loaded by default; you cannot choose it until you load it.　CENTER2 默认未加载，必须先 Load。
2. In the Layer Manager, assign CENTER2 to layer AXIS.　在图层管理器中把 CENTER2 指定给 AXIS 图层。
3. LTSCALE ↲ → type 40 ↲.　键入 LTSCALE，输入 40。

2.5　图层的五种状态　Five layer states

State 状态	Visible 可见	Selectable 可选	Editable 可改	Use 用途
Current 当前	√	√	√	New objects land here　新绘对象归入此层
On 打开	√	√	√	Normal state　常规状态

State 状态	Visible 可见	Selectable 可选	Editable 可改	Use 用途
Off 关闭	×	√	√	Hidden, still drawable　隐藏但仍可绘制
Frozen 冻结	×	×	×	Fully isolated　完全隔离
Locked 锁定	√	√	×	Trim around it without damaging it　可修剪而不破坏

2.6　改正画错图层的对象　Moving an object to the right layer

PROPERTIES ↲ （CTRL + 1）　Modify → Properties　修改 → 特性

1. Select the object or objects.　选中对象。
2. In the Properties palette, open the Layer list and choose the correct layer.　在特性面板的 Layer 下拉列表中选择正确的图层。
3. Press Esc to deselect.　按 Esc 取消选择。

3　轴网　Axis Grid

The axis grid is the only thing that makes a construction drawing measurable.

3.1　The first axis line

LINE ↲　Draw → Line　绘图 → 直线

1. Make AXIS the current layer.　把 AXIS 置为当前图层。
2. Turn F8 on.　打开 F8 正交。
3. LINE ↲ — click a start point, move the cursor to the right, type 30000 ↲, then Esc.

※ The line must be longer than the building so the axes extend past the outline.

3.2　偏移　Offset

OFFSET ↲ （O ↲）　Modify → Offset　修改 → 偏移

1. OFFSET ↲.
2. Type the distance, e.g. 2600 ↲.　键入偏移距离。
3. Click the line to be offset.　用拾取框点选待偏移的直线。
4. Click the side to offset towards.　用十字光标指定偏移方向。
5. The command stays active and the distance stays in effect. Press Esc before typing a new distance.

3.3　建立轴网　Building the grid

Do one direction completely, then the other. The table below gives the cumulative distance from the first axis,

4　墙体　Walls

A wall is drawn as a multiline: a single object made of two parallel elements, one on each side of the axis. A 240 wall sits 120 to either side of its axis.　墙体用多线绘制：一个由两条平行图元构成的单一对象。240 墙以轴线为中，两侧各 120。

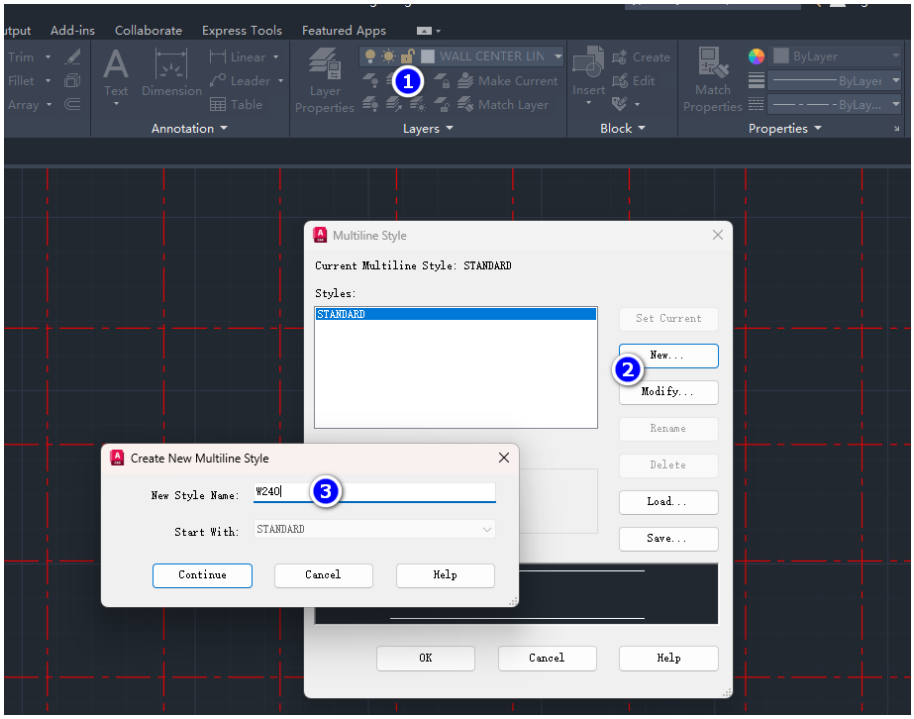
4.1　多线样式　Multiline style W240

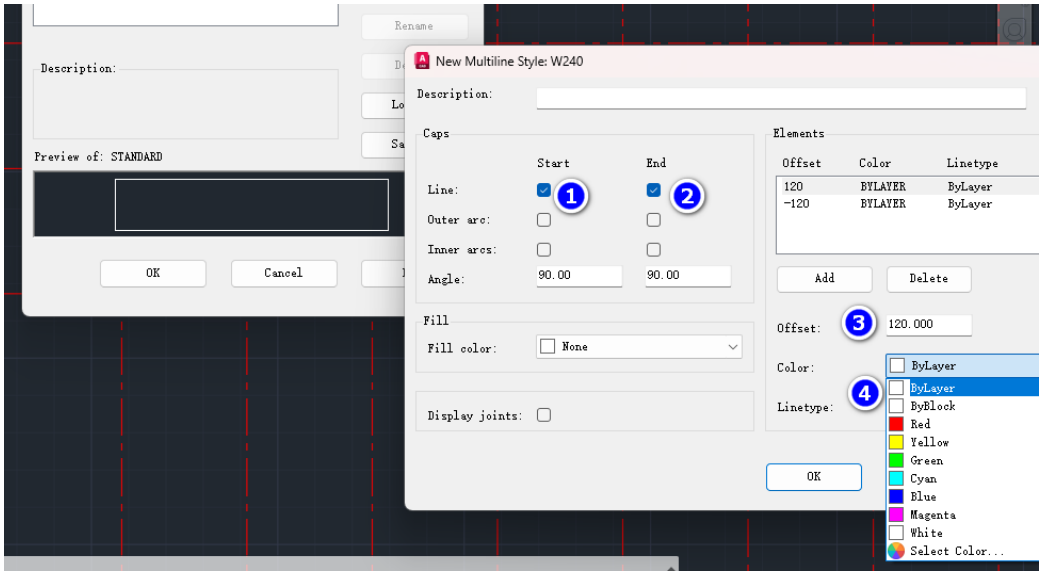
MLSTYLE ↲　Format → Multiline Style　格式 → 多线样式

1. MLSTYLE ↲ → New… → name it W240 → Continue.　新建并命名 W240（以墙厚命名便于查找）。

2. Element Properties → Add → set offset to 120.　在图元中添加偏移 120。
3. Add → set offset to −120.　再添加偏移 −120。
4. Set the colour and linetype of both elements to ByLayer.　两条图元的颜色与线型设为 ByLayer。
5. OK → Set Current → OK.
6. save the multi line style in the system.

※ The offsets are measured from the axis, one to each side: +120 and −120, not 240.





4.2　绘制墙体　Drawing the walls

MLINE ↲　(ML ↲)　Draw → Multiline　绘图 → 多线

Before the first click, check the three current settings shown on the command line:　动笔前先核对命令行显示的三项当前值：

Setting 设置	Value 取值	Key 子选项	Meaning 含义
Justification 对正	Zero	J ↲	The multiline is centred on the points you click 以所点的点为中

Setting 设置	Value 取值	Key 子选项	Meaning 含义
Scale 比例	1	S ↵	Element offsets are used as entered 图元偏移按原值使用
Style 样式	W240	ST ↵	The style you just defined 刚定义的样式

Interior walls　内墙

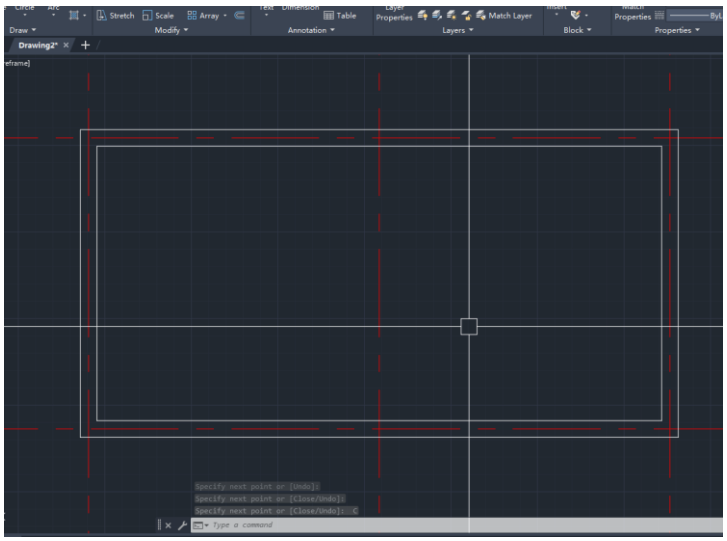
1. Keep WALL current. Snap to axis intersections only — nothing else.　保持 WALL 为当前层，只捕捉轴线交点。
2. Draw each interior wall run as one multiline, from axis intersection to axis intersection.　每道内墙用一条多线一次画完，端点落在轴线交点上。

4.3　编辑交点　Editing the joints

MLEDIT ↵　Modify → Object → Multiline　修改 → 对象 → 多线

Tool 工具	中文	Use 用途
Cross Merge	十字合并	Two walls crossing — both pass through 两墙交叉贯通
Cross Close	十字闭合	Two walls crossing — one is closed off 交叉处封闭
Tee Merge	T 形合并	One wall meets another and stops 一墙止于另一墙
Tee Close	T 形闭合	T junction closed off T 形交接处封闭

1. MLEDIT ↵ and choose the tool that matches the junction.　键入 MLEDIT，选择与交接形式对应的工具。
2. For a Tee: click the wall that stops first, then the wall that runs through.　T 形合并：先点“被打断”的墙，再点“贯通”的墙。
3. Work through every junction in the plan.　逐个处理全部交点。



4.4　修剪　Trimming

TRIM ↵　(TR ↵)　Modify → Trim　修改 → 修剪

1. TRIM ↵, then press ↵ again — every visible object becomes a cutting edge.　键入 TRIM 后直接回车，全部可见对象即为边界。
2. Click the segments you want removed.　点击要删除的线段。
3. Hold Shift and click to extend instead of trim.　按住 Shift 点击则为延伸。
4. Esc to finish.　按 Esc 结束。

5 填充 Hatching

5.1 What gets hatched

Drawing 图种	Cut through 被剖切	Hatched 需填充
Plan 平面	a horizontal cut 水平剖切	structural columns, shear walls 结构柱、剪力墙
Section 剖面	a vertical cut 竖向剖切	beams, slabs, columns 梁、板、柱
Elevation 立面	nothing 无剖切	wall finishes — material, not solid 饰面材质

5.2 填充操作 Applying a hatch

HATCH ↵ (BHATCH ↵ / H ↵) Draw → Hatch 绘图 → 图案填充

1. Set the correct layer first — COLUMN before hatching columns, PAVING before hatching floors. 先置为正确的图层。
2. HATCH ↵ → choose the pattern
3. Pick Points — click inside the enclosed area and AutoCAD searches for the boundary. Select Objects — pick the boundary objects yourself. 拾取点：点击封闭区域内部；选择对象：自行点选边界。
4. Adjust the scale, then ↵ to finish. 调整比例后回车结束。

6 查询自检 Self-check by Measuring

MEASUREGEOM ↵ (MEA ↵) Tools → Inquiry 工具 → 查询

Option 选项	Key	Measures 测量对象	Required check 必测项
Distance	D ↵	distance between two points 两点距离	Overall width 建筑总宽 = 24600
Area	AR ↵	area of a closed region 闭合区域面积	One interior bay 任意一跨
Angle	A ↵	angle between two lines 两线夹角	Wall corners = 90°

附录 A　命令索引　Command Index

Command 命令	中文	Use 用途	§
OPTIONS	选项	Crosshair size, colour theme, pickbox	1.1 / 1.2
DSETTINGS (DS)	草图设置	Object snap — four snaps only	1.3
LAYER (LA)	图层	Create and manage the seven layers	2.1
LINETYPE	线型	Load CENTER2 and assign it to AXIS	2.3
LTSCALE	线型比例	Set to 40 so the dashes read	2.3
REGEN	重生成	Refresh the display after LTSCALE	2.3
LWEIGHT	线宽	Turn the lineweight display on	2.4
PROPERTIES (Ctrl+1)	特性	Move an object to the correct layer	2.6
LINE (L)	直线	The first axis line	3.1
OFFSET (O)	偏移	Every other axis, by typed distance	3.2
MLSTYLE	多线样式	Define W240 with +120 / −120	4.1
MLINE (ML)	多线	Draw the walls	4.2
MLEDIT	多线编辑	Clean the wall junctions	4.3
TRIM (TR)	修剪	Remove unwanted segments	4.4
HATCH (H / BHATCH)	图案填充	SOLID for columns, ANSI31 for paving	5.2
MEASUREGEOM (MEA)	查询	Distance, area, angle — the final check	6

Troubleshooting

Symptom 现象	Cause 原因	Fix 处理
Lines are not exactly horizontal　线不水平	F8 was off　F8 未开	Turn F8 on, redraw the line
The axis line looks continuous, not dashed　轴线看起来是实线	LTSCALE = 1, or CENTER2 not loaded　线型比例或线型未加载	§2.3 — LTSCALE 40, then REGEN
The wall runs to one side of the axis　墙偏在轴线一侧	Justification ≠ Zero　对正方式不是 Zero	J ↵ → Zero, redraw
TRIM will not cut the wall　TRIM 剪不动墙	A multiline is a single object　多线是单一对象	§4.3 — use MLEDIT
MLEDIT trimmed the wrong wall　剪错了墙	Picked in the wrong order　点选顺序颠倒	Undo; pick the stopping wall first
Hatch fails, or floods the whole drawing　填充失败或溢出全图	Gap in the boundary　边界有缺口	§5.2 — zoom in and close the gap
The hatch is a grey smear　填充成一片灰	Pattern scale too small　填充比例过小	Increase the scale; judge it on the print
A command will not start　命令无法启动	Another command is still running　上一个命令未结束	Press Esc twice, then retype